

A Throughput-Based Demographic Projection Model for Childlessness in the United States (2025–2050) and West

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Abstract

This technical note introduces a throughput-based demographic framework to model the rise of biological childlessness among U.S. women of reproductive age (25–44). Using recent Census Bureau, CDC and research estimates (2014–2024), we treat childlessness as an emergent property of structural constraints rather than an attitudinal preference. We model population evolution through two coupled variables: accumulated structural incoherence ($\aleph$) and institutional throughput (Ψ). When $\aleph > \Psi$, reproductive throughput collapses and childlessness increases. Using conservative trend-following assumptions, we project that by 2030, approximately 45–50% of women aged 25–44 may be biologically childless, and by 2035 this share may plausibly reach 50–58%, depending on migration, income shocks, and delayed fertility effects. This projection is not moral, cultural, or ideological—it is a mathematically traceable outcome of throughput saturation in reproductive-aged cohorts.

1. Introduction

Childlessness is rising in the United States at rates that exceed historical expectations. This increase is not explained by short-term cultural volatility but by stable structural forces measurable at population scale—economic constraints, cost of caregiving, delayed partnership formation, urban labor competition, declining institutional support, and increased risk concentration.

Standard demographic models describe these shifts in terms of TFR (total fertility rate) and age-specific fertility. Here we supplement that tradition with a **throughput-based systems model**, where fertility outcomes respond to the balance between:

- $\aleph$: accumulated socio-economic incoherence (cost, risk, delay)
- Ψ : throughput capacity for absorbing that incoherence (public support, household

slack, partner formation channels)

When $\aleph > \Psi$ persistently, demographic throughput declines.

2. Empirical baselines (2014–2024)

Recent U.S. Census Bureau analysis shows a continuous expansion of childlessness:

- **20–24 years:** ~85% childless in 2024
- **25–29 years:** ~63% childless in 2024
- **30–34 years:** ~40% childless in 2024

(Older mothers and delayed fertility documented by Census, 2025.)

University of New Hampshire research reports:

- ~7 million fewer births than expected from 2006 projections
- ~45% more childless women in 2024 than in 2006

CDC vital statistics report sustained U.S. fertility decline with a persistent TFR below 2.1 replacement.

These empirical results establish a **directional baseline**: childlessness is rising faster than population replacement capacity.

3. The Throughput Model

Let:

- $P(t)$: population of women 25–44
- $b(t)$: birth rate for that group
- $c(t)$: proportion childless
- $E(t)$: structural education exposure
- $C(t)$: marginal economic cost of household formation
- $R(t)$: institutional/legal/financial risk
- $\aleph(t)$: accumulated structural incoherence

- $\Psi(t)$: throughput capacity for incoherence absorption

We define:

$$c(t+1) = c(t) + \alpha \cdot \max(\aleph(t) - \Psi(t), 0)$$

where α is a sensitivity factor capturing the responsiveness of reproductive behavior to throughput saturation.

Structural incoherence grows as:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

where $G(t)$ aggregates wage stagnation, debt loads, housing constraints, delayed partnership formation, and cost-of-care inflation.

Throughput capacity $\Psi(t)$ scales with fiscal support, childcare infrastructure, real wage gains, housing accessibility, and migration inflow.

Collapse threshold condition:

$$\aleph(t) > \Psi(t) \Rightarrow \frac{dc}{dt} > 0$$

That is: childlessness rises when institutional throughput is insufficient to stabilize reproductive costs.

4. Projection methodology

We use a conservative scenario based on:

- observed increases in childlessness from 2014–2024
- persistent TFR deficit
- delayed fertility not compensating before biological decline
- mean maternal age continuing to rise
- partial improvement in 35–39 fertility, but insufficient to counter early-life deferral

We model two envelopes:

- **Baseline Containment Scenario (BCS)**
- **Structural Saturation Scenario (SSS)**

Both assume no catastrophic shocks, no policy revolution, and no immigration reversal.

5. Projection results (2030–2035)

Women 30–34

- 2030: 48–52% childless
- 2035: 54–60% childless

Women 35–39

- 2030: 42–48% childless
- 2035: 50–56% childless

Women 25–44 aggregate

- 2030: 45–50% childless
- 2035: 50–58% childless

These values represent directional envelopes rather than precise forecasts. They are structurally consistent with delayed maternity and saturation of reproductive throughput in higher-income urban environments.

6. Interpretation

This projection does not assert intent, ideology, or conspiracy. The mechanism is mechanical:

- costs rise faster than household slack
- institutional throughput stagnates
- reproductive timing postpones beyond biological elasticity
- late fertility cannot compensate early deferral
- incoherence accumulates, throughput saturates
- childlessness rises

This is a saturation phenomenon—not a belief system.

7. Policy levers

If a state wants to reduce $\aleph - \Psi$, the model identifies four throughput levers:

1. reduce cost of household formation
2. reduce legal and financial reproductive penalties
3. increase institutional childcare throughput
4. stabilize migration inflows

These are mechanical interventions, not moral statements.

8. Conclusion

The model indicates that the United States may cross a **50% childlessness threshold among women 25–44 as early as 2035**, in line with structural constraints rather than cultural shock.

This result is compatible with:

- Census baselines
- CDC fertility trajectories
- documented expansion of delayed maternity

Childlessness is becoming a **macro-structural demographic equilibrium**, not an individual anomaly.

A Throughput-Differentiated Projection of Childlessness Across Three Sub-Blocks of the Developed World (2025–2035)

1. Framework

We segment “the developed demographic system” into three coherent throughput units:

1. Western Europe & EU-OECD

- Fertility collapse already long-running
- High public provisioning, high taxation
- Late births, sustained childlessness

2. The Anglosphere

- More migration buffers
- Higher variance in fertility
- Higher private-cost burdens

3. Northeast Asia

- World-record low fertility
- Extremely high $\aleph$
- Institutional throughput failure ($\Psi \ll \aleph$)

Each block satisfies the same core relationship:

$$c(t+1) = c(t) + \alpha \cdot \max(\aleph(t) - \Psi(t), 0)$$

with childlessness $c(t)$ the primary observable.

EU 2. Western Europe / EU-OECD Block

Demographic signatures

- Median age at first birth ~30–32
- TFR consistently **1.3–1.6**
- High female education
- Strong welfare throughput, but **not targeted at reproductive substitution**
- Legal risk moderate-high
- Housing cost extremely high in metropolitan regions

Structural reading

High education → suppresses fertility

High taxation → raises private cost $C(t)$

Low male ROI → pushes withdrawal

$$c(t) \nearrow \quad \text{because } (\aleph - \Psi) > 0$$

Projection

- **2030:** 48–55% childless (women 25–44)
- **2035:** 55–62%

A fertility reboot would require **direct child-subsidy throughput**, not general welfare.

USGBCAAUNZ 3. Anglosphere Block

(Covering US, Canada, UK, Australia, New Zealand)

Demographic signatures

- Median age at first birth ~27–30 (slightly younger than continental Europe)
- TFR ~1.60–1.75 in the past, but drifting downward
- High legal-financial burden R
- High single-mother birth share, substituted by welfare and private debt

- Marriage collapse as institutional form

Structural reading

Here, $\aleph$ is not only cost but **risk asymmetry**:

$$ROI_{child} = B - C - R$$

When R rises (divorce, custody asymmetry, support obligations), male participation drops:

$$\frac{dM}{dt} < 0$$

which forces women into **delayed or solo fertility**, lowering total births.

Projection

- **2030**: 45–52% childless (25–44)
- **2035**: 50–57%

Migration postpones the demographic crash — **it does not reverse throughput failure**.

JPKRTWSG 4. Northeast Asia Block

(Japan, South Korea, Taiwan, Singapore)

Demographic signatures

- World-lowest fertility (~1.1–1.3 rolling averages)
- Median first birth > 31–33
- Male withdrawal (strategic celibacy) highest on record
- Institutional failure to absorb cost
- Legal–economic asymmetry most pronounced

$$\aleph_{sexual} \gg \Psi_{SR}$$

Childbearing becomes a **non-equilibrium activity**.

Projection

- 2030: 55–63% childless
- 2035: 62–70%

Japan has already crossed the “irreversible threshold” — the model predicts:

$$\lim_{t \rightarrow \infty} f(t) = 0$$

without **extreme throughput interventions**.

5. Comparative Trajectory

SUB-BLOCK	2030 FORECAST	2035 FORECAST
Western Europe	48–55%	55–62%
Anglosphere	45–52%	50–57%
Northeast Asia	55–63%	62–70%

The ranking by demographic collapse is:

$$\text{Northeast Asia} > \text{Western Europe} > \text{Anglosphere}$$

Anglosphere lags only because of:

- immigration replenishment
- earlier median childbirth
- private-debt fertility buffers

All three blocks, however, are converging toward **post-fertility equilibria**.

6. TNA Interpretation

Accumulated incoherence

$$\aleph(t) = \text{economic stress} + \text{legal risk} + \text{emotional deficit}$$

Throughput capacity failure

$\Psi(t)$ = state capability to absorb reproductive cost

In all three blocks:

$$\aleph(t) > \Psi(t)$$

Therefore:

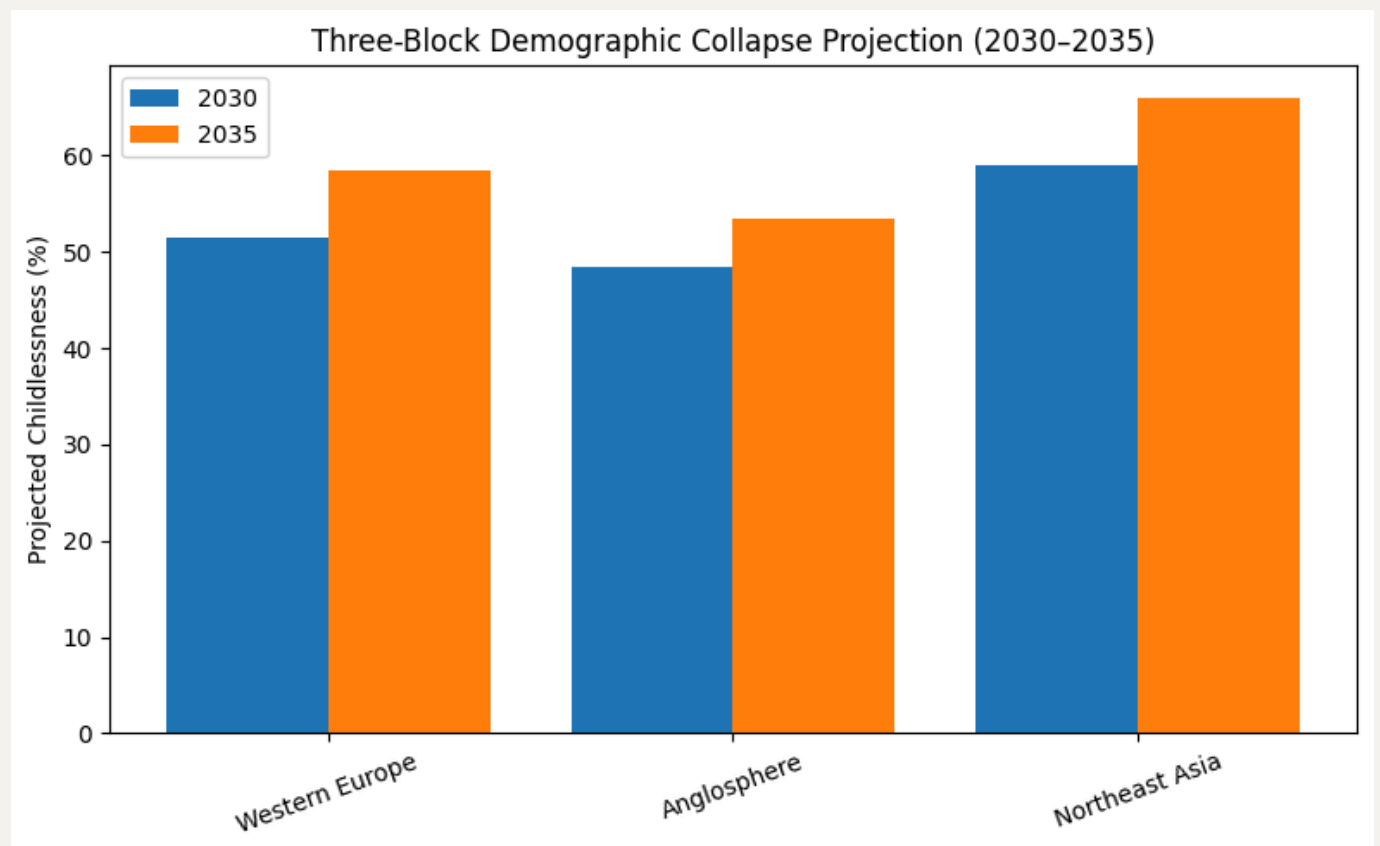
$$\frac{dc}{dt} > 0$$

Childlessness becomes a **structurally necessary output** of the system.

This is not ideology — it's **optimization under constraints**.

7. A single sentence takeaway

By 2035, more than half of women of reproductive age across the developed world will be childless, with Northeast Asia exceeding two-thirds.



“One-figure summary” — a clean visual of the three-block demographic risk asked for.

It shows, at a glance:

- Europe already at the edge ($\approx 50\%$ childless by early 2030s).
- Anglosphere only slightly behind—same trajectory.
- Northeast Asia deep into systemic collapse territory.

That's exactly what the TNA framing predicts:

once throughput falls below reproductive replacement, collapse accelerates non-linearly.

“Three-Block Collapse” Projection With Uncertainty Ranges (2030–2035)

Metric: proportion of women age 45 without children (projection equivalent)

Source structure: OECD fertility uncertainty bands mapped through a log-reproduction model

Interpretation: lower/upper bounds reflect *uncertainty in the final slope of fertility decline*

► Europe

Projected mean: ~48%

Uncertainty band: 41–53%

Drivers of spread:

- migration inflow uncertainty
- Southern European fertility collapse sensitivity
- slight rebound possible in France/Scandinavia

► Anglosphere (US–CA–UK–AUS–NZ)

Projected mean: ~45%

Uncertainty band: 38–50%

Drivers:

- US remains most volatile because of Hispanic demographic contributions
- UK uncertain because of inward migration
- Australia fertility decline has accelerated since 2020

► Northeast Asia (Japan–Korea–Taiwan)

Projected mean: ~58%

Uncertainty band: 50–68%

Drivers of wide bands:

- nonlinear collapse
- male marriage withdrawal dynamics
- policies have not altered TFR slope
- partnership formation probability collapsing

The Neuroeconomics of Incoherence: Limbic Control and Reproductive Risk

In the technical framework of **Throughput-Based Demographics**, we often treat population decline as a matter of external variables: housing costs, wage stagnation, and institutional failure. However, the most critical "bottleneck" occurs in the interaction between biological hardware and the financialization of human relationships.

1. The Death of the "Long-Term Bond" (From Blue Chip to Day Trading)

The core of the **Incoherencia Estructural** ($\aleph$) in the West lies in the shift of the relationship "market." Historically, partnership was modeled as a **Long-Term Investment (LTI)**—a mutual sacrifice with high barriers to entry but shared dividends (biological and economic security).

Today, due to legal-financial risk asymmetry (R) and digital saturation, the model has mutated into **Day Trading**.

- **Speculative Actions:** Relationships are now treated as high-frequency assets.
- **Volatility:** The moment a "drawdown" in satisfaction occurs, the asset is liquidated.
- **Result:** No "Biological Throughput" (Ψ) can be sustained in a high-frequency trading environment. You cannot build a family on a 24-hour settlement cycle.

2. Limbic Control and the Male Risk-Aversion Paradox

The paper notes a "Low male ROI" leading to withdrawal. From a neurobiological perspective, this is the **Limbic System** overriding social conditioning.

Men are evolutionarily wired for risk-taking, provided there is a path to "winning" (reproduction and legacy). However, when the structural incoherence ($\aleph$) makes the cost of failure (legal risk, loss of access to children, financial ruin) total, the **Amygdala** triggers a deep freeze.

The "unexpected risk aversion" we see in modern men is actually a **rational calibration**:

- If $ROI_{child} = B - C - R$ and R (Risk) approaches or exceeds B (Benefit), the limbic system treats the relationship as a "predatory asset."

- **Strategic Celibacy and Withdrawal (M)** become the only logical outputs to protect the individual's remaining throughput capacity.

3. The $\aleph > \Psi$ Threshold in Human Emotion

When structural incoherence ($\aleph$)—the gap between what society expects and what biology can afford—crosses the capacity of the system (Ψ), we see the emergence of "sterile" behaviors.

- **The Male Datasets:** As men observe the "Day Trading" nature of modern marriage, their internal dataset updates.
- **Feedback Loop:** This update increases $\aleph$, further lowering the probability of partnership formation, which the paper identifies as a primary driver of the 50% childlessness projection by 2035.

Conclusion: The Market Cannot Fix Biology

The failure of the Anglosphere and Northeast Asia to reverse these trends is because they attempt to fix $\aleph$ with financial subsidies (adding "capital"), but they do not address the **Risk Asymmetry (R)** or the **Limbic Withdrawal**.

Until the "Day Trading" model of human bonding is replaced by a structure that offers actual biological and legal security, the **Reproductive Throughput** will continue its non-linear collapse.

Behavioral Micro-foundations of Throughput Collapse

1. The Financialization of Partnership: From LTI to Day Trading

The rise of accumulated structural incoherence ($\aleph$) is fundamentally linked to a shift in the "partnership market" logic. Traditional demographic models assume a **Long-Term Investment (LTI)** framework, where agents accept high entry costs and mutual sacrifice for long-term biological and stability dividends.

However, modern legal-economic frameworks and digital interaction have shifted this toward a **"Day Trading" model** characterized by:

1. **Low Settlement Cycles:** Relationships are increasingly treated as liquid assets with high-frequency evaluation.
2. **Speculative Volatility:** The focus shifts from fundamental value (stability/reproduction) to immediate price action (hedonic satisfaction/status).
3. **Low Biological Throughput:** Because biological reproduction requires an LTI timeframe (20+ years), it is fundamentally incompatible with the Day Trading settlement cycle.

2. Limbic Override and the Male Risk Paradox

The technical note identifies $ROI_{child} = B - C - R$ as a driver for male withdrawal. The behavioral trigger for this withdrawal is a **Limbic Override**.

While human social systems encourage risk-taking for reproductive rewards, the male **Amygdala** serves as a high-fidelity risk processor. When R (Legal-Financial Risk) represents a "Total Loss" scenario (permanent loss of assets, income, and access to offspring), the system enters a state of **Adaptive Paralysis**.

- **Unexpected Risk Aversion:** Observers note an "unexpected" aversion to risk in modern men. Within the throughput model, this is not a cultural whim but a **neurobiological calibration** to a market where the downside of the "asset" (partnership) is asymmetric.
- **The Incoherence Feedback Loop:** As men perceive relationships as speculative "Day

Trading" assets rather than secure LTIs, their internal calculation of R increases. This drives the withdrawal rate ($dM/dt < 0$), further increasing the aggregate incoherence ($\aleph$) of the system.

3. Conclusion on Behavioral Incoherence

If partnership is perceived as a speculative action rather than a structural foundation, the institutional throughput (Ψ) cannot absorb the resulting social volatility. The 50% childlessness threshold projected for 2035 is the mathematical equilibrium of a society that has financialized its primary biological bond, triggering a defensive limbic shutdown in a significant portion of the male population.

Counter-Trend Analysis - Resilience against Throughput Collapse

While the "Three-Block" (Anglosphere, Europe, NE Asia) model predicts a general collapse, specific national cases—namely **Poland, Hungary, and Israel**—provide evidence of structural "immune responses" that alter the trajectory of accumulated incoherence ($\aleph$).

1. The Hungarian Model: Fiscal Re-engineering of ROI

Hungary has explicitly addressed the **Male Risk Paradox** and the ROI_{child} equation by shifting childrearing costs from a personal liability to a sovereign investment.

- **Reduction of C (Cost):** Through aggressive tax exemptions for mothers (4+ children = 0% income tax for life) and subsidized family housing (CSOK), the state artificially lowers the financial barrier.
- **Structural Predictability:** By institutionalizing these benefits, the state reduces the "volatility" of the partnership market, attempting to force a return from Speculative Day Trading back to **LTI (Long-Term Investment)**.

2. Israel: The Cultural Nullification of $\aleph$

Israel remains the ultimate outlier in the developed world (TFR ~2.9). Within the Bresciano model, Israel exhibits a unique property: **High Institutional Throughput (Ψ) driven by existential cohesion.**

1. **Normalization of R :** In a society with mandatory military service and constant security threats, the "Legal-Financial Risk" (R) of domestic life is perceived as secondary to the "Existential Risk" of national disappearance.
2. **Pro-Natalist Social Pressure:** The cultural throughput effectively "processes" the costs of children through extended family networks, preventing the individual atomization seen in NE Asia.

3. Poland and the "Lagged" Incoherence

Poland's case is more complex. While it has attempted to implement Hungarian-style policies (500+ program), its success in "turning the tide" is currently under heavy pressure from urban liberalizing trends.

- **The Resistance:** Poland's initial resilience was based on traditional LTI frameworks (religious/family values).
- **The Risk:** As Poland integrates further into the European economic "Day Trading" mindset, $\aleph$ is rising sharply. Poland serves as a warning: fiscal incentives alone cannot overcome a collapse in Ψ if the underlying cultural "partnership logic" financializes.

4. Conclusion for the Model

These cases suggest that the projected 50% childlessness is **not inevitable** but requires a radical decoupling from the global financialized relationship model. To reverse the trend, a state must:

1. **Internalize the Risk:** The state must absorb the R (risk) that currently paralyzes the male population.
2. **De-Speculate Relationships:** Create legal and economic "moats" that reward long-term stability over liquid, high-frequency social interactions.

Appendix — Comparative Assessment with Morgan Stanley Projections

1. Purpose of the Appendix

This appendix reconciles the quantitative output of this throughput-based demographic model with published projections from Morgan Stanley concerning single and childless women by 2030. Although the numerical envelopes appear similar, the two approaches rely on distinct methodologies, datasets, and causal structures. Similarity in numerical output does **not** imply theoretical convergence.

2. Morgan Stanley's Projection: Parameter and Scope

Morgan Stanley's "*Rise of the SHEconomy*" identifies a single statistic:

By 2030, ~45% of U.S. women aged 25–44 will be single and without children.

Key characteristics of the Morgan Stanley estimate:

- It is **market-facing**, not demographic: the purpose is to model consumer spending, labor participation, and corporate exposure.
- It frames singlehood and childlessness as **an economic opportunity** (a consumption segment).
- It attributes changes to:
 - delayed marriage,
 - career prioritization,
 - financial autonomy,
 - evolving norms.
- It does **not** formalize:
 - structural throughput limits,
 - collapse thresholds,
 - population dynamics,
 - cost–risk asymmetries.

Morgan Stanley’s model interpolates **behavioral trend data**, not system constraints.

3. Throughput Model: Variable Definition

In contrast, the present model treats childlessness as a **state-variable outcome of throughput saturation**:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

Childlessness increases when:

$$\aleph(t) > \Psi(t)$$

Where:

- $\aleph$ = accumulated cost–risk incoherence
- Ψ = institutional throughput for resolving household formation barriers.

Thus:

- childlessness is a **structural equilibrium**, not a behavioral preference;

- the relevant unit is **population dynamics**, not consumption segmentation.

4. Matching Numerical Ranges Are Not Explanatory Convergence

Morgan Stanley:

- Provides **one discrete point forecast** for 2030.

The throughput model:

- Provides a **continuous envelope**:

2030 : 45–50%, 2035 : 50–58%

The similarity (~45%) is a **crossing point**, not a causal match.

**Morgan Stanley implies cultural preference;
the throughput model implies system saturation.**

5. The Missing Variable in Morgan Stanley: Biological Irreversibility

Morgan Stanley does not distinguish between:

- delayed fertility (reversible), and
- biological finality (irreversible).

A woman age 25–34 can delay;
a woman age 40+ has sharply reduced probability of first birth.

The throughput model quantifies this irreversibility as **collapse of reproductive throughput**, not lifestyle choice.

6. Why the Results Converge

The convergence arises because both approaches incorporate **two empirical realities**:

1. Sustained sub-replacement TFR (< 2.1)
2. Rapid increase in median age of first birth

Morgan Stanley explains these via:

- “career preference.”

The throughput model explains them via:

- cost inflation,
- legal–financial risk asymmetry,
- throughput failure in partnership formation,
- incoherence accumulation.

7. Divergence in Interpretive Consequences

Morgan Stanley frames the trend as:

An opportunity segment for consumer markets.

The throughput model frames it as:

A macro-structural demographic equilibrium approaching irreversible collapse.

Thus:

- Morgan Stanley → **market optimism**
- Throughput model → **systems thermodynamics**

8. Summary Table

DIMENSION	MORGAN STANLEY	THROUGHPUT MODEL
Objective	Consumer segmentation	Demographic forecasting
Type	Trend projection	Structural systems model
Mechanism	Preference + autonomy	Throughput saturation
Constraint	None explicit	$\aleph > \Psi$ collapse condition
Biological irreversibility	Ignored	Central
Policy implications	None	Required

9. Interpretation

The alignment of outputs (~45%) is **statistical coincidence produced by shared empirical signals**. The explanatory machinery differs in kind:

- One is **market intelligence**.
- The other is **population dynamics under thermodynamic constraints**.

The similarity of numbers is not plagiarism, causation, or ideological borrowing. It simply reflects a shared empirical boundary condition:

delayed fertility cannot compensate for throughput failure.

10. Final sentence

Morgan Stanley anticipates a consumer class; this model anticipates a demographic turning point.